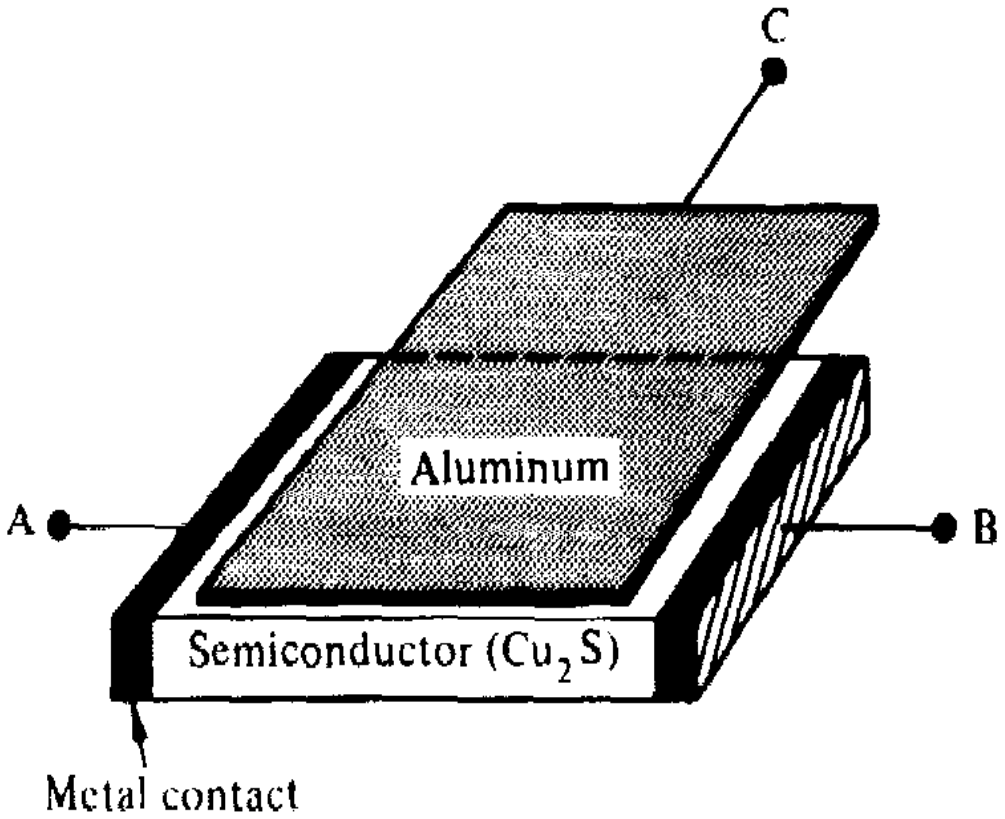
MOS Capacitor

- Introduction
- Ideal MOS Capacitor
  - Qualitative Description of MOS Capacitor (CAP)
  - Quantitative Analysis of MOS Capacitor (CAP)
- *C*-*V* Curves of MOS Capacitor
- Non Ideal MOS Capacitor
- Applications of *C*-*V* Curve

FET (Field Effect Transistor)

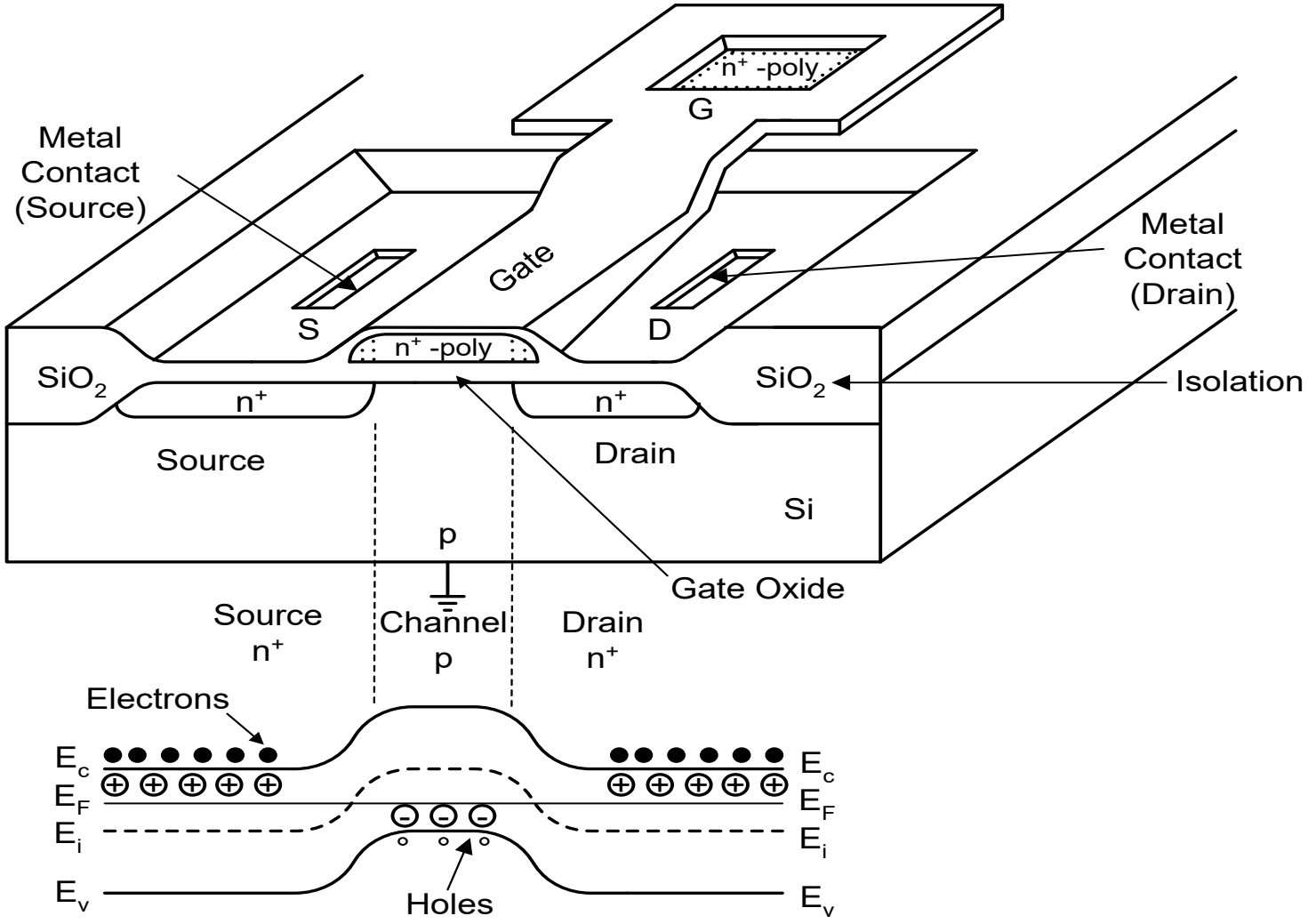


- Basic concept was introduced by J. E. Lilienfeld in 1926.
- Modulate the conductance and current flowing in semiconductor between A and B by applied voltage at C.
- Normal electric field to the surface of semiconductor controls the conductivity → concept of FET.

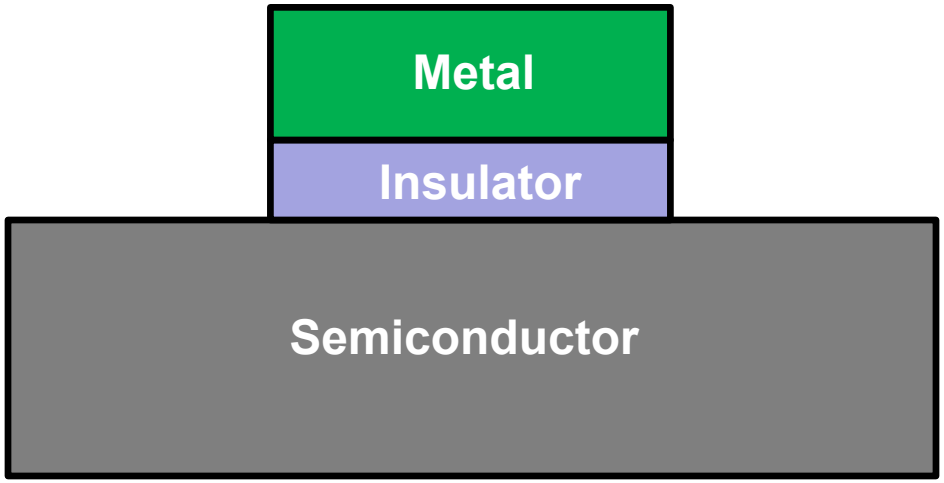
Historical Background

- BJT was introduced in industry in 1948.
- W. Shockley introduced and developed the first modern FET, J-FET in 1950s.
- In 1960, Kahng in Bell Lab. demonstrated MOSFET (Metal Oxide Semiconductor Field Effect Transistor).
- Commercial MOSFET was demonstrated by Fairchild Semiconductor and RCA in 1964.
- From 1980s, MOSFET began to dominate the IC industry with CMOS technology.

MOSFET (Metal Oxide Semiconductor Field Effect Transistor)



MOS Capacitor



- Good simpler structure to understand “field effect” device.
- Gate-Oxide-Si substrate stack in a 1-D view : MOSFET without source and drain.

- Widely used as the test structure for characterizing GOI (Gate Oxide Integrity), reliability of gate oxide film, and other properties (interface quality between Si and gate oxide, contamination of fab line, etc)
- One of the most fundamental circuit elements devices in modern VLSI.

MOS Capacitor

- Introduction
- Ideal MOS-capacitor
  - **Qualitative Description of MOS CAP**
  - Quantitative analysis of MOS CAP
- C-V Curves of MOS-capacitor
- Non Ideal MOS-capacitor
- Application of C-V Curve

Assumptions for Ideal MOS Capacitor

- Energy difference between metal work function, Φ_M and the semiconductor work function, Φ_s is zero, i.e., $\Phi_{MS} = 0$.

  → Flat band condition when $V_G = 0$.

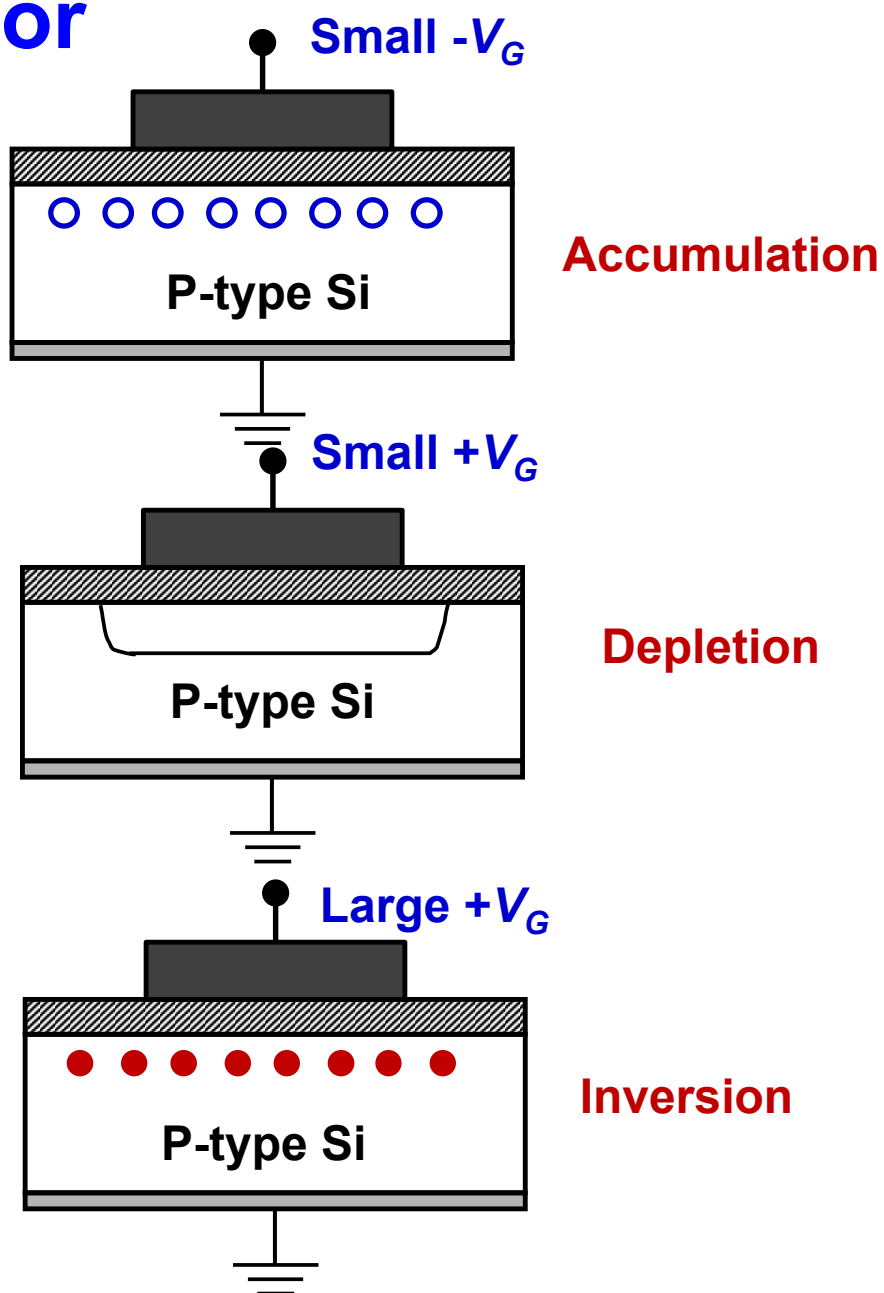
- The only existing charges in the structure are those in the semiconductor and those with the equal but opposite sign on the metal surface adjacent to the insulator.
- No carrier transport through the insulator under DC biasing conditions, i.e., the resistivity of the insulator is infinity.

  → no mobile charge carriers nor any charge centers exist.
- Semiconductor region is uniformly doped.
- Gate layer is considered to be an equipotential region.
- Approximations:

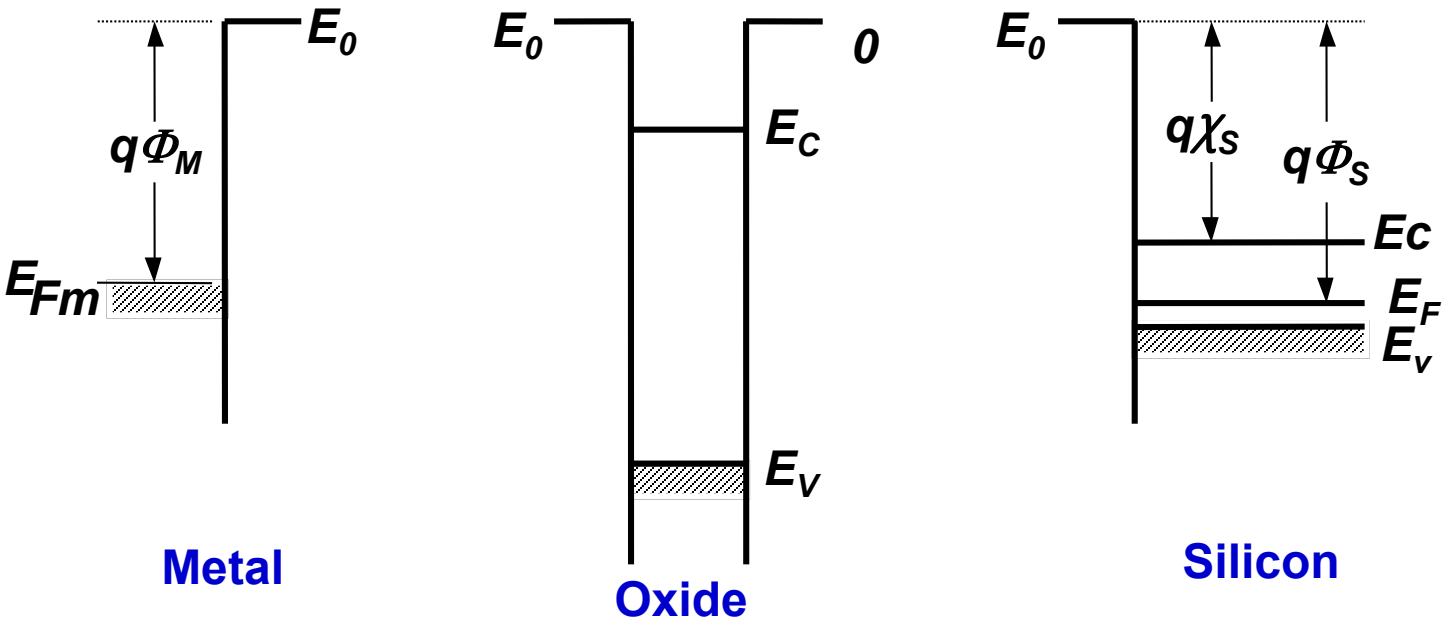
  1. Maxwell-Boltzman Approximation
  2. Complete Ionization

Qualitative Description of MOS Capacitor Behavior

- 3 operation modes w.r.t. applied gate voltage, V_G.
- Depletion, Inversion, and Accumulation.
- Modulation of semiconductor surface state with different polarity and magnitude of V_G.

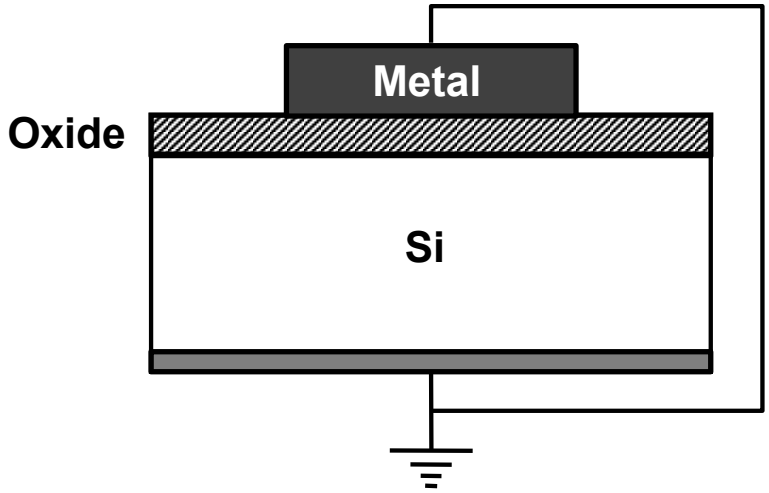


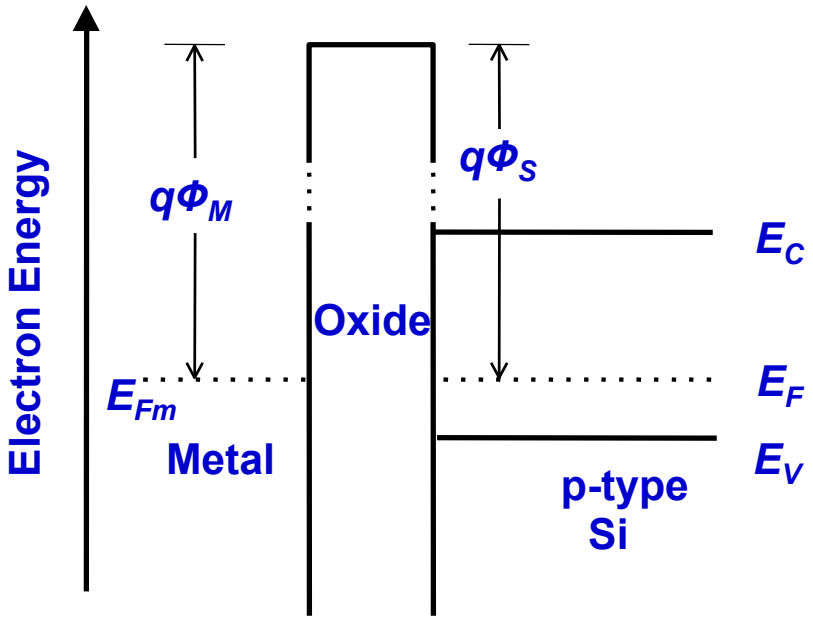
Energy Band Diagram Representation of MOS capacitor



- E_0 : Vacuum level
- Φ_M : Metal work function
- Φ_S : Semiconductor work function
- χ_S : Electron affinity of semiconductor

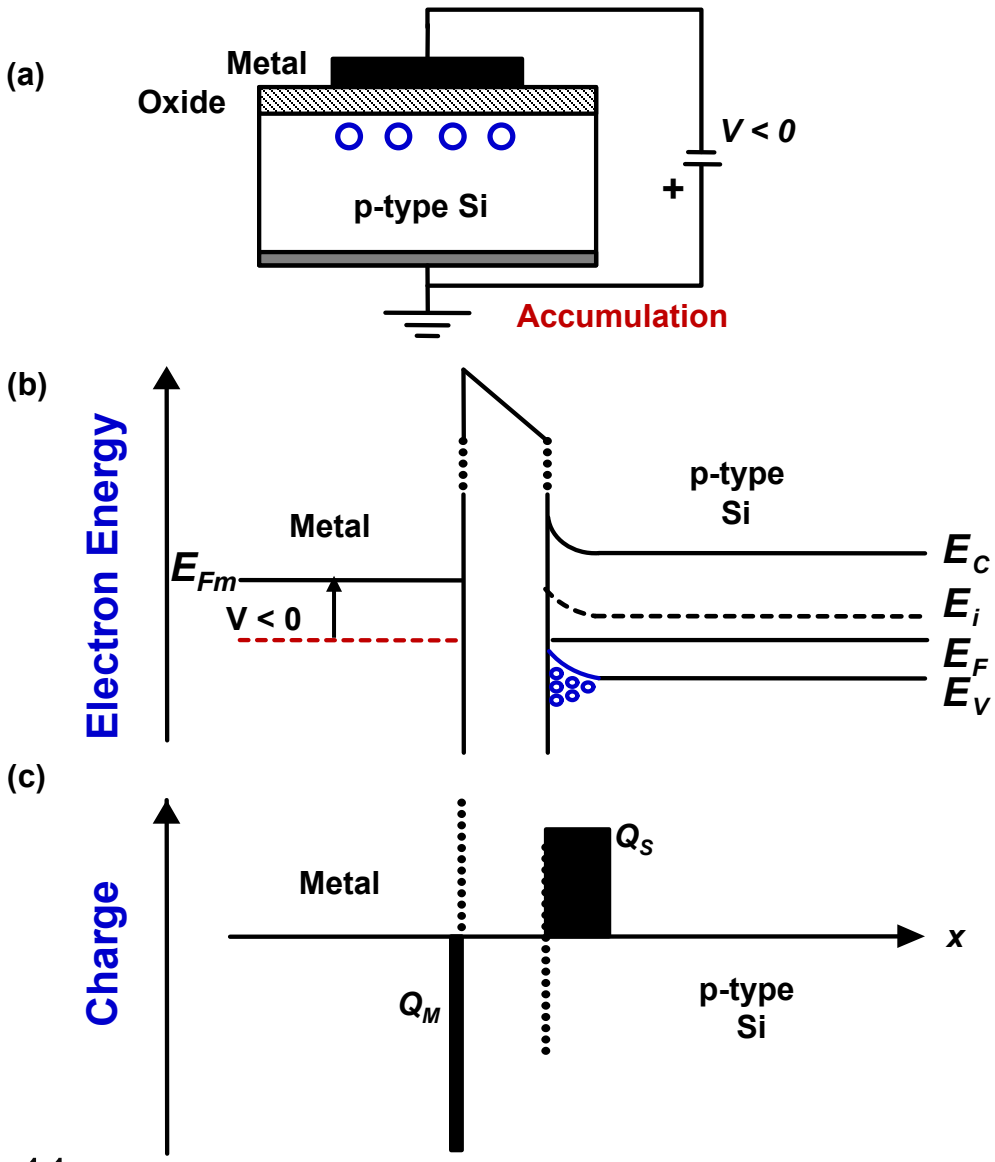
Flat Band Condition (Ideal MOS Cap)





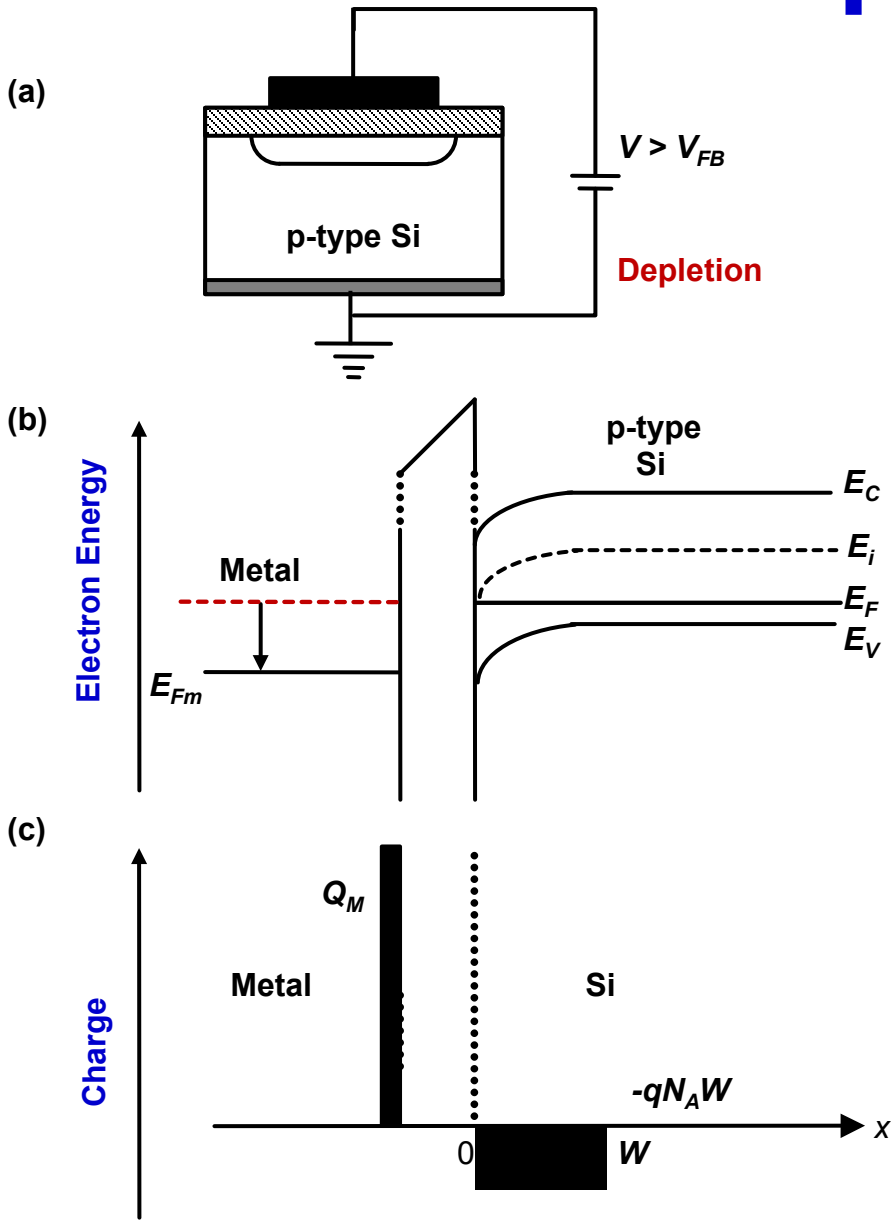
- At $V_G = V_{FB}$, the Fermi level is constant throughout the structure → no current flow.
- Since Φ_{MS}=0, V_{FB}=0V.
- Energy bands are flat within Si, no substrate charge, Si is neutral → $Q_s = 0$.
- Energy bands in the oxide are also flat and horizontal.

Accumulation



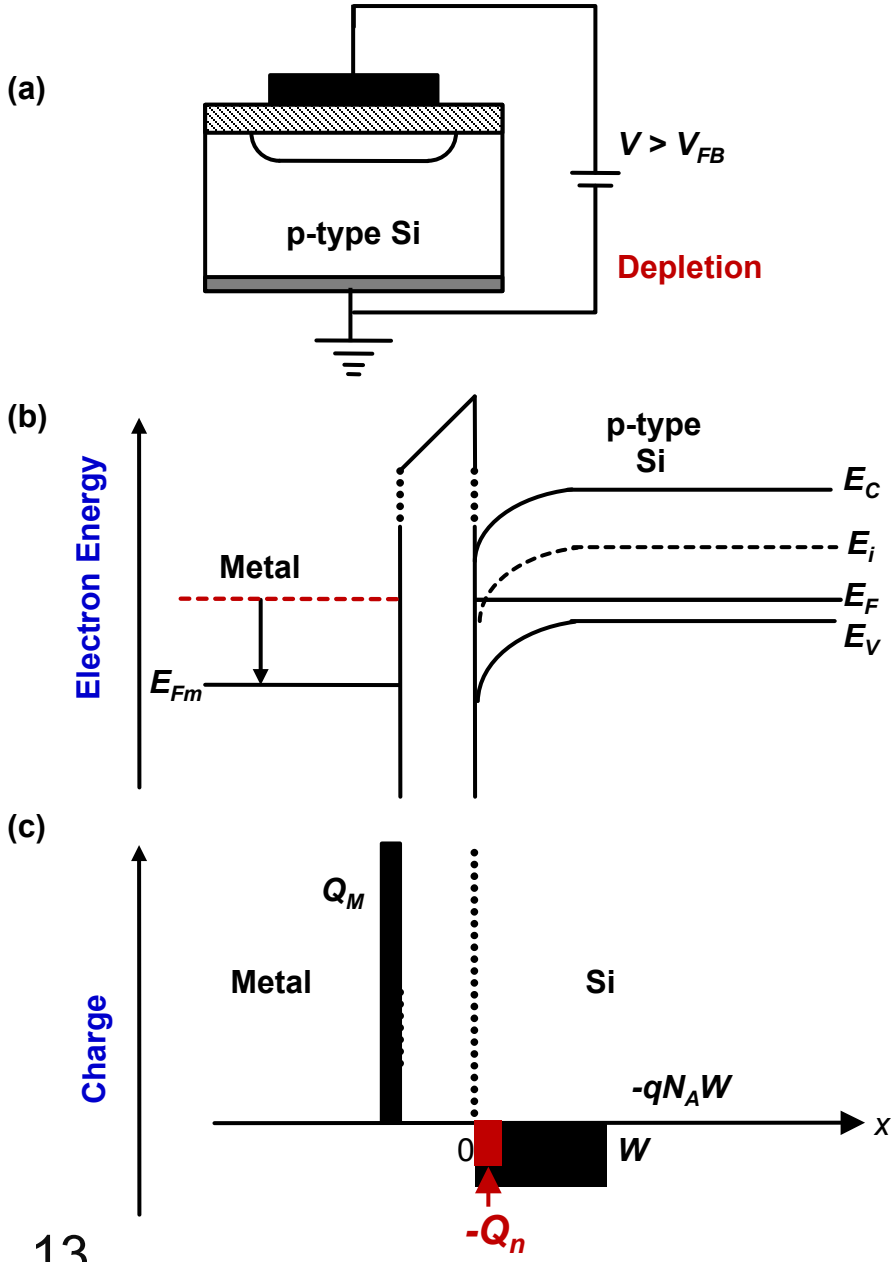
- Negative V_G for p-type substrate.
- Negative charge on the metal is balanced by equal and positive charge on the silicon surface.
- Fermi level of the metal is shifted upwards with respect to the Fermi level in the silicon.
- Voltage is dropped mostly across the oxide but also some voltage dropped across the silicon → slope in the energy bands in both oxide and the bands in the silicon surface to bend.
- Fermi-level in the silicon is still flat, because there is no current flowing in the silicon. There is the oxide layer which blocks all mobile charge movement.

Depletion



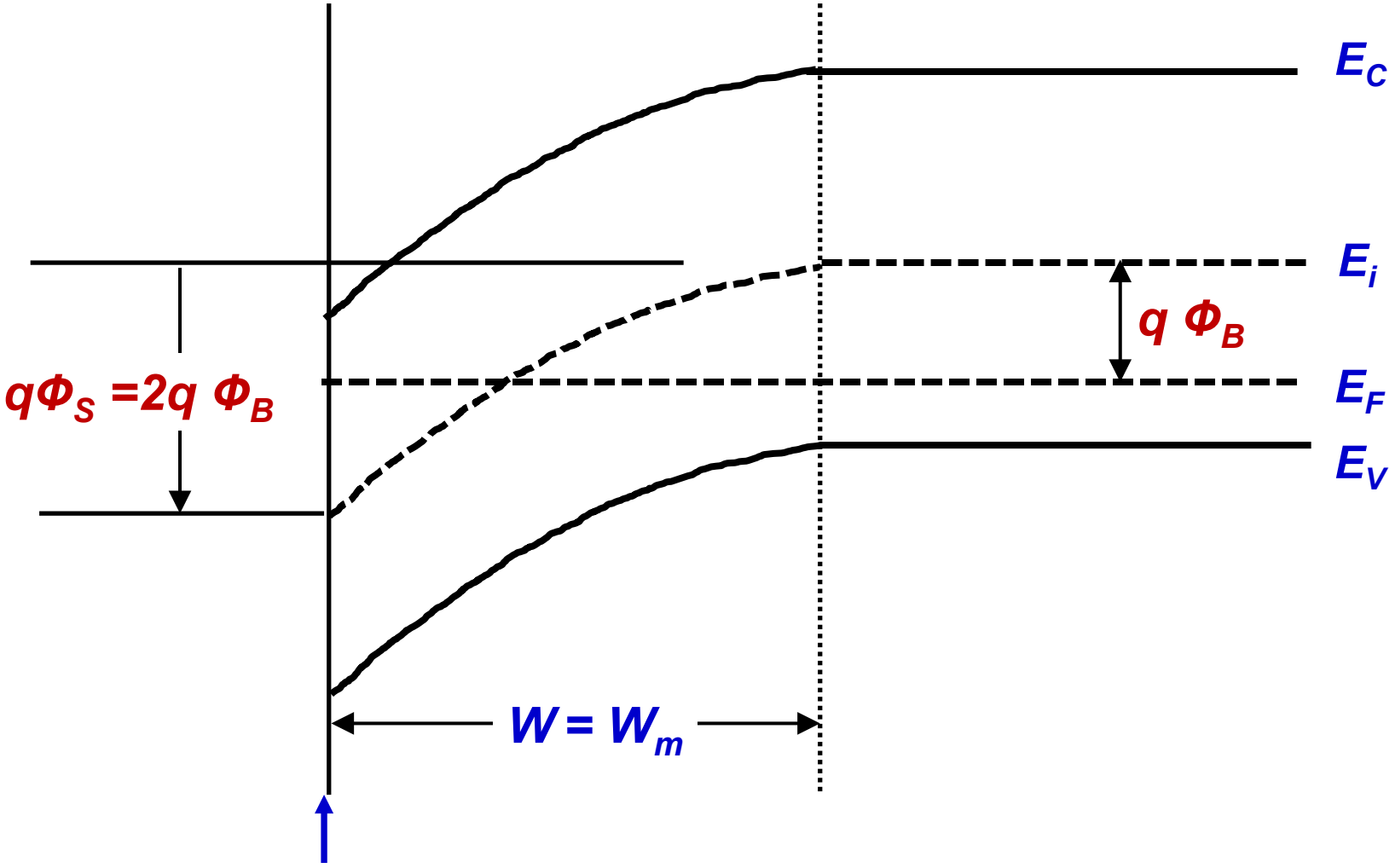
- Positive V_G for p-type substrate.
- Fermi level of the metal is shifted downwards, and silicon surface is also bent downwards, with flat Fermi-level in the silicon.
- Holes will move away from the surface of the silicon resulting in a depletion region being formed in the surface: consistent with the band bending.
- Total amount of positive charge in the metal Q_M is balanced exactly by the total amount of negative charge in the depletion layer charge, given by $q \cdot N_A \cdot W$ per unit area (where W is the width of the depletion region).

Weak Inversion

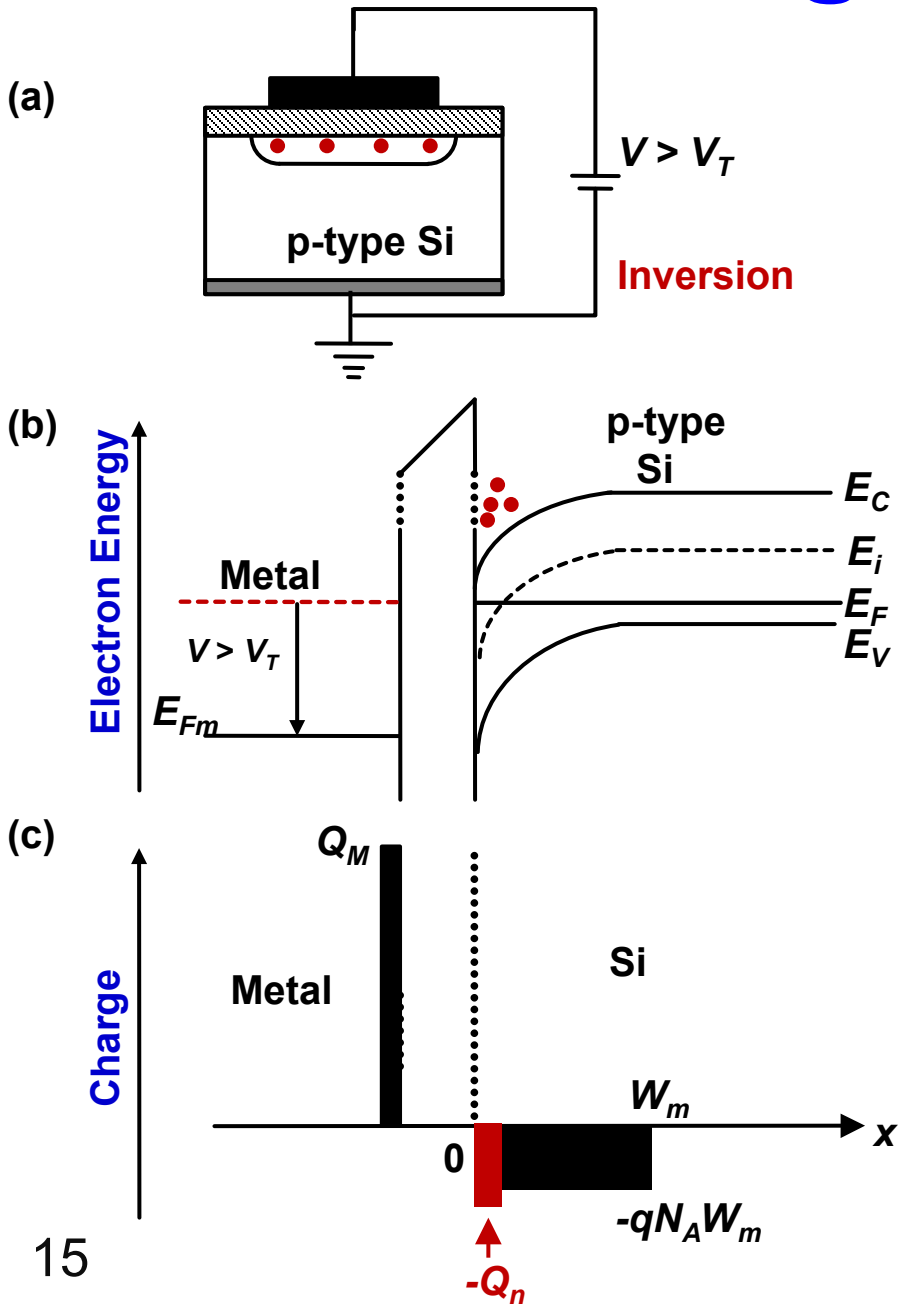


- Positive V_G for p-type substrate.
- Fermi level of the metal is shifted downwards, and silicon surface is also bent downwards, with flat Fermi-level in the silicon.
- Holes will move away from the surface of the silicon resulting in a depletion region being formed in the surface: consistent with the band bending.
- At the onset of weak inversion, electron concentration is higher than the intrinsic electron concentration, but smaller than ionized acceptors. Continue to increase in W.

Onset of Strong Inversion



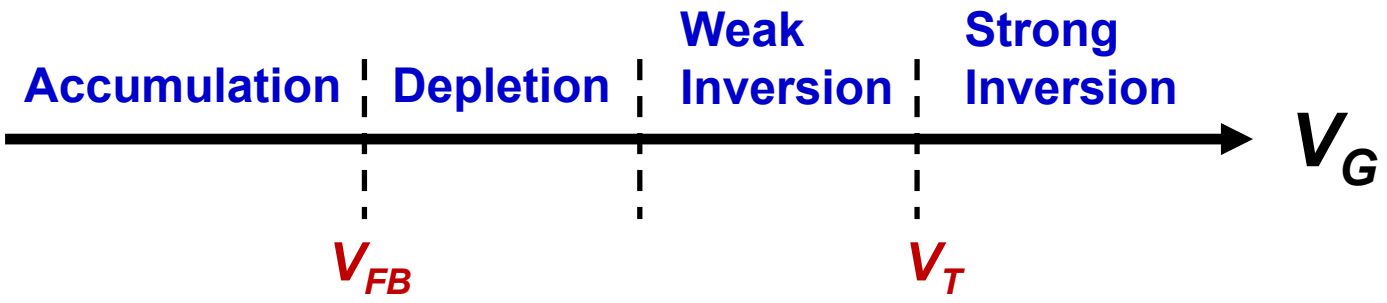
Strong Inversion



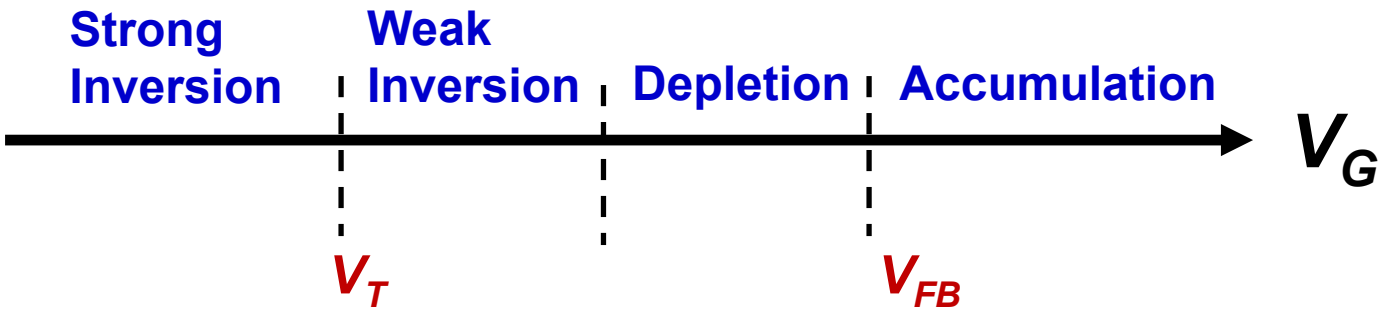
- Increasing the positive V_G, more positive charge is placed on the gate, requiring more negative charges to be induced at the silicon surface by increasing W.
- The depletion layer width will increase until it reaches a maximum value, W_M.
- At the onset of strong inversion, the additional induced negative charge to be supplied by mobile electron at the surface more than by ionized acceptors. No increase in W.
- In the inversion condition, more electrons than holes at the silicon surface, and the surface is said to be **inverted, i.e. n-type and not p-type**.

Summary

MOS structure with P-Type Substrate:



MOS structure with N-Type Substrate:



Note: V_{FB} = 0 V when Φ_{MS} = 0 V.